

K-means Clustering: Algorithm and Analysis

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1 Introduction

Let $x_1, \dots, x_N \in \mathbb{R}^d$ be a finite dataset. In clustering, the objective is to partition these points into a fixed number of groups without access to labels or responses.

K-means is a classical clustering method based on minimizing within-cluster squared Euclidean distances. Despite its algorithmic simplicity, the underlying optimization problem is nonconvex and admits multiple stationary solutions. This note presents the algorithm, its exact objective, and the basic theoretical guarantees it satisfies.

2 Model and Problem Formulation

We seek to represent the dataset using K points in $\mathbb{R}^d$. Each data point is associated with exactly one representative, and the quality of this representation is measured by the squared Euclidean distance between a point and its associated representative.

To express the association between data points and representatives explicitly, we introduce binary assignment variables. For each data point x_i and each representative index k , the variable r_{ik} indicates whether x_i is assigned to representative k .

Fix an integer $K \geq 1$. Formally, define the assignment set

$$\mathcal{R} = \left\{ (r_{ik}) \in \{0, 1\}^{N \times K} : \sum_{k=1}^K r_{ik} = 1 \quad \forall i \right\}.$$

Let $\mu_1, \dots, \mu_K \in \mathbb{R}^d$ denote the representative points. Given assignments $r \in \mathcal{R}$ and representatives $\mu_1, \dots, \mu_K$, define the objective

$$J(r, \mu) = \sum_{i=1}^N \sum_{k=1}^K r_{ik} \|x_i - \mu_k\|^2.$$

The K-means problem is the joint minimization of $J(r, \mu)$ over $r \in \mathcal{R}$ and $\mu \in (\mathbb{R}^d)^K$.

Algorithm 1 Abstract Lloyd Iteration

- 1: Initialize $\mu_1, \dots, \mu_K \in \mathbb{R}^d$
 - 2: **repeat**
 - 3: $r \leftarrow \arg \min_{r' \in \mathcal{R}} J(r', \mu)$
 - 4: $\mu \leftarrow \arg \min_{\mu' \in (\mathbb{R}^d)^K} J(r, \mu')$
 - 5: **until** convergence
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Each subproblem admits a closed-form solution, stated below.

3 Analysis

Throughout this section, assume that no cluster is empty and that no distance ties occur.

Theorem 1 (Mean minimizes squared distance). *Let $x_1, \dots, x_n \in \mathbb{R}^d$ and define*

$$f(\mu) = \sum_{i=1}^n \|x_i - \mu\|^2.$$

Then f has a unique minimizer

$$\mu^* = \frac{1}{n} \sum_{i=1}^n x_i.$$

Proof. The function f is a strictly convex quadratic. Setting $\nabla f(\mu) = 0$ yields $n\mu = \sum_{i=1}^n x_i$, which uniquely determines μ^* . $\square$

Theorem 2 (Optimal assignment). *Fix $\mu_1, \dots, \mu_K \in \mathbb{R}^d$. An assignment $r^* \in \mathcal{R}$ minimizes $J(r, \mu)$ if and only if, for all i ,*

$$r_{ik}^* = 1 \iff k \in \arg \min_{1 \leq j \leq K} \|x_i - \mu_j\|^2.$$

Proof. For fixed μ , the objective separates across i . Each term is minimized by assigning x_i to its nearest centroid. $\square$

Theorem 3 (Finite termination of Lloyd's algorithm). *Assume no empty clusters and no distance ties. Let $(r^{(t)}, \mu^{(t)})$ be generated by alternating minimization of J . Then there exists $T < \infty$ such that*

$$(r^{(t)}, \mu^{(t)}) = (r^{(T)}, \mu^{(T)}) \quad \forall t \geq T.$$

Proof. Whenever the assignment changes, the objective strictly decreases. Since $\mathcal{R}$ is finite and J is bounded below, only finitely many assignment changes are possible. $\square$

Combining the theorems in this section yields the standard implementation of the algorithm. See Algorithm 2 below.

Algorithm 2 Lloyd's Algorithm for K-means

Require: Data $x_1, \dots, x_N \in \mathbb{R}^d$, number of clusters K

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1: Initialize  $\mu_1, \dots, \mu_K \in \mathbb{R}^d$ 
2: Initialize  $r^{(0)} \in \mathcal{R}$  arbitrarily
3:  $t \leftarrow 0$ 
4: repeat
5:   for  $i = 1$  to  $N$  do
6:      $r_i^{(t+1)} \leftarrow \arg \min_{1 \leq k \leq K} \|x_i - \mu_k\|^2$ 
7:   end for
8:   for  $k = 1$  to  $K$  do
9:      $\mu_k \leftarrow \frac{\sum_{i=1}^N \mathbf{1}\{r_i^{(t+1)} = k\} x_i}{\sum_{i=1}^N \mathbf{1}\{r_i^{(t+1)} = k\}}$ 
10:  end for
11:   $t \leftarrow t + 1$ 
12: until  $r^{(t)} = r^{(t-1)}$ 
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4 Engineering Considerations

The analysis above relies on idealized assumptions that are rarely met in practical settings. Implementations of K-means therefore incorporate additional design choices. These choices affect robustness and efficiency but do not alter the underlying objective.

Empty clusters. During iteration, it is possible for a representative to receive no assigned data points. In this case, the centroid update is undefined. A common remedy is to reinitialize the representative, for example by selecting a data point uniformly at random or by perturbing an existing representative. This operation is heuristic and may increase the objective, but restores a valid state from which optimization can continue. Empty clusters typically indicate poor initialization or an overly large choice of K .

Termination criteria. Theoretical analysis assumes termination when assignments stabilize. In practice, convergence is often detected using approximate criteria, such as a bound on the change in centroids, a bound on the decrease of the objective, or a fixed iteration budget. These criteria reflect a trade-off between solution quality and runtime, and are necessary when K-means is used as a subroutine in larger systems.

Redundant representatives. It is common for two representatives to become close or nearly identical, particularly when K is large relative to the intrinsic structure of the data. Such redundancy leads to unstable assignments and sensitivity to small perturbations. This phenomenon reflects the symmetry of the objective and the lack of identifiability of cluster labels, rather than numerical error. In high dimensions, redundant representatives are the norm rather than the exception.

Acceleration techniques. The assignment step typically dominates the computational cost, scaling as $O(NK)$ per iteration. Practical implementations exploit structure to reduce this cost, including vectorized distance computations, triangle inequality pruning, and approximate nearest-neighbor search. These techniques preserve the objective but change the algorithmic profile, often yielding substantial speedups in large-scale settings.

Choice of K . The K-means objective is monotone decreasing in K , and therefore does not determine an optimal number of clusters. Selecting K is an external modeling decision. Heuristic methods such as the elbow plot examine the marginal decrease of the objective as K increases, providing qualitative guidance. Such methods are descriptive and should not be interpreted as optimization principles.

5 Applications

K-means is rarely used as a final model. Its primary value lies in exploratory analysis, compression, and initialization of more structured methods. This section summarizes representative use cases and their limitations.

Image segmentation. Pixels are embedded as vectors encoding intensity, color, or local features and clustered in feature space. The resulting segmentation produces a piecewise-constant approximation of the image. This approach is computationally efficient and effective for images with large

homogeneous regions, but it ignores spatial continuity and struggles with textured or fine-grained structure. Consequently, K-means is typically used as a baseline or as an initialization for spatially regularized models.

Customer and asset segmentation. In marketing and finance, entities are represented by feature vectors summarizing behavior, preferences, or return characteristics. K-means produces coarse groupings that are easy to interpret and visualize. However, results are highly sensitive to feature scaling and to the choice of distance metric. The method is best viewed as a descriptive tool rather than a principled statistical model of heterogeneity.

Handwritten digit data. Applied to image vectors, K-means clusters capture low-level structure such as stroke density and orientation. While clusters do not align reliably with semantic labels, they often reveal meaningful geometric organization of the data. K-means is therefore useful for visualization, dimensionality reduction, or as a preprocessing step for supervised learning, but not as a standalone classifier.

Use as a subroutine. K-means is frequently used to initialize more expressive models, including Gaussian mixture models, vector quantization schemes, and dictionary learning algorithms. In these settings, its role is algorithmic rather than inferential: it provides a reasonable starting point that accelerates convergence and improves stability of subsequent optimization procedures.

Remark 1 (Game-theoretic interpretation). *The function $-J$ defines an exact potential game between an assignment player and a centroid player. Lloyd’s algorithm coincides with best-response dynamics, and its fixed points correspond to pure-strategy Nash equilibria of this game.*